

Priyansh Vaghela

Phone: (208)-366-9658 | pgvaghela20@gmail.com | github.com/pgvaghela | linkedin.com/in/priyanshvaghela
No sponsorship required (Green Card)

EDUCATION

University of Arizona

Tucson, AZ

B.S. in Computer Science, Minors: Entrepreneurship & Innovation, Info Sci & Tech

Aug. 2022 – May 2026

- **Awards:** Dean's List (Spring 2026, 4× total); Academic Year Academic Distinction (2023–2024, 2× total)
- **Relevant Courses:** Object-Oriented Programming, Systems Programming, Data Structures, Algorithms, Machine Learning

SKILLS

Programming Languages: Python, TypeScript, JavaScript, Java, C# (.NET), C++, C, Swift, SQL, HTML/CSS

Frameworks & Libraries: React, FastAPI, Spring Boot, Angular, ASP.NET Core, Tailwind, PyTorch

Data & Telemetry: PostgreSQL, MySQL, pandas, NumPy, scikit-learn, TF-IDF, data pipelines

Cloud & DevOps: AWS (EC2, RDS, S3, CloudWatch), Docker, Azure DevOps, GitHub Actions, Git, Linux/Unix

Testing & CI/CD: Unit Testing (Jest, JUnit, PyTest), Integration Testing, Postman, CI/CD (GitHub Actions, Azure Pipelines)

Methodologies & Design: Agile/Scrum, SaaS, SDLC, RESTful API design, design patterns

EXPERIENCE

AI/Data Analyst Intern | Python, PyTorch, pandas, boto3, Paramiko, AWS S3, Gemini API

May. 2025 – Aug. 2025

Quantara AI

Remote

- Built zero-disk SFTP-to-S3 ingestion streaming 2M+ rows of Zywave cyber-risk data through BytesIO without local writes
- Engineered resilient CSV parser handling malformed rows, mixed delimiters, and encoding failures in vendor feeds
- Shipped Gemini labeling script tagging vulnerability descriptions with loss type and MITRE ATT&CK techniques
- Added row-level checkpointing and 60s retry-on-rate-limit so multi-hour Gemini jobs resume from last row on failure
- Trained PyTorch embedding-pool classifier on 6-class loss-type labels with class weighting to handle long-tail imbalance
- Generated 1,400+ row synthetic training set across AWS Inspector, CrowdStrike, Tenable, and Sentinel schemas

Software Engineering Intern | Angular, C# .NET, Azure DevOps, Git, Postman, Swagger

May. 2024 – Aug. 2024

Micron Technology

Boise, ID

- Built full stack Angular and C# .NET application used by 200+ daily users, integrating 10+ REST endpoints
- Implemented 12 REST APIs with Swagger/OpenAPI docs; validated through Postman, resolved 20+ defects in sprint QA
- Led UAT triage with stakeholders; resolved 50+ issues and clarified requirements to unblock sign-off for production
- Delivered sprint features across US, Taiwan, and India teams, coordinating handoffs and validation in 2-week cycles

Teaching Assistant | Python, Tutoring

Jan. 2024 – Dec. 2025

University of Arizona

Tucson, AZ

- Led weekly CS 110 lab sessions of 30+ students, combining short lectures with hands-on Python programming exercises
- Held 4 office hours weekly, graded quizzes and exams, and debugged students' Python code one-on-one during sessions

PROJECTS

Job Application Agent | Python, FastAPI, Claude API, React, TypeScript, PostgreSQL, Docker

Apr. 2026 – May 2026

- Built agentic loop with Claude API tool use, web search, and iterative refinement over resume and cover letter drafts
- Cut per-iteration token cost 90% via prompt caching on system prompt and tool definitions across multi-turn agent loops
- Added pgvector RAG over resume, projects, and past writing; agent retrieves top-k chunks per JD via cosine similarity
- Wrapped sync web search in run_in_executor to keep FastAPI's async event loop unblocked during agent tool calls
- Built React + TypeScript replay dashboard with Overview, Resume Diff, Cover Letter, and Agent Trace tabs for debugging

F1 Dashboard | Angular 19, C# / ASP.NET Core 9, PostgreSQL, ML.NET, Python, Docker

May 2026 – Jun. 2026

- Built full-stack Formula 1 analytics platform deployed across Vercel, Render, and Neon with multi-source data ingestion
- Diagnosed OOM kills on 512MB Render container loading full races into pandas, moved ingest off-platform to fix it
- Trained ML.NET FastTree winner predictor with leak-free walk-forward validation after catching in-sample memorization
- Built Python/FastF1 telemetry pipeline ingesting 160K+ samples per race with SVG coordinate normalization for replay

Stock Trading Simulator | C++17, Crow, SQLite, AWS C++ SDK, CloudWatch, React, Docker

Jun. 2025 – Jul. 2025

- Rewrote backend in C++17 with Crow WebSocket server and SQLite repos using prepared statements with RETURNING
- Built mutex-synchronized matching engine with priority_queue buy and sell books, persisting trades on each match
- Polled Alpha Vantage for 5 tickers on 60s background thread with 5-min cache, rate-limit fallback, and WebSocket fan-out
- Published OrdersPerMinute to CloudWatch via AWS C++ SDK behind a HAVE_AWS_SDK ifdef with stdout fallback

ClearView News Dashboard | Python, FastAPI, React, TypeScript, PostgreSQL

Feb. 2026 – Apr. 2026

- Built full-stack dashboard (FastAPI + React/TS) aggregating 50-100 daily articles with bias labels and source links
- Engineered ingestion pipeline pulling ~75 articles/day from 8 outlets via NewsAPI with cross-source deduplication
- Designed TF-IDF clustering pipeline grouping articles by topic so the same story across outlets shows as one cluster